Stage 2: Caterpillar–Plant–Bee Model

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1 Model

$$\begin{aligned}
\frac{dP}{dt} &= rP\left(1-\frac{P}{K}\right) - aPC + \beta\frac{vPB}{H+P}\left(1-\frac{P}{K}\right),\\
\frac{dC}{dt} &= \varepsilon aPC - \mu C,\\
\frac{dB}{dt} &= \gamma\frac{vPB}{H+P} - dB - qB^2.
\end{aligned} \tag{1}$$

1.1 Assumptions

1. Plant growth, caterpillar feeding, and caterpillar growth and mortality follow the Stage 1 model.

2. Flower visitation per bee increases with plant abundance and saturates at v, reaching half its maximum when $P = H$.

3. Flower visits support plant growth and bee population growth. Additional plant growth from bees is limited by the factor $1 - P/K$.

4. Bees experience natural mortality and density-dependent competition.

1.2 Variables and Units

Variable	Meaning	Units
P	Plant abundance	Plants
C	Caterpillar abundance	Caterpillars
B	Bee abundance	Bees
t	Time	Days

1.3 Parameters and Units

Parameter	Biological meaning	Units
r	Intrinsic plant growth rate	day^{-1}
K	Plant carrying capacity	plants
a	Rate of plant loss due to caterpillar feeding	$(\text{caterpillar} \cdot \text{day})^{-1}$
ε	Efficiency of converting plant losses into caterpillar population growth	caterpillars/plant
μ	Natural caterpillar mortality rate	day^{-1}
v	Maximum flower visits per bee	$\text{visits}/(\text{bee} \cdot \text{day})$
H	Plant abundance at which visitation per bee reaches half its maximum	plants
β	Additional plant growth per bee flower visit	plants/visit
γ	Bee population growth contribution per flower visit	bees/visit
d	Natural bee mortality rate	day^{-1}
q	Strength of density-dependent bee competition	$(\text{bee} \cdot \text{day})^{-1}$

2 Equilibrium Solutions

Equilibrium name	**Equilibrium** (P^*, C^*, B^*)	**Biologically feasible when**	**Stability**
Extinction (E_0)	$(0, 0, 0)$	Always	Unstable saddle
Plants only (E_P)	$(K, 0, 0)$	Always	Stable if $K < \dfrac{\mu}{\varepsilon a}$ and $\dfrac{\gamma v K}{H + K} < d$
Plant–bee (E_{PB})	$\left(K, 0, \frac{1}{q}\left(\frac{\gamma v K}{H+K} - d\right)\right)$	$\dfrac{\gamma v K}{H + K} > d$	Stable if $K < \dfrac{\mu}{\varepsilon a}$
Plant–caterpillar (E_{PC})	$\left(\frac{\mu}{\varepsilon a}, \frac{r}{a}\left(1 - \frac{\mu}{\varepsilon a K}\right), 0\right)$	$K > \dfrac{\mu}{\varepsilon a}$	Stable if $\dfrac{\gamma v P^*}{H + P^*} < d$
Coexistence (E_{PCB})	$P^* = \dfrac{\mu}{\varepsilon a}$, $B^* = \dfrac{1}{q}\left(\dfrac{\gamma v P^*}{H + P^*} - d\right)$, $C^* = \dfrac{1}{a}\left(1 - \dfrac{P^*}{K}\right) \times \left(r + \dfrac{\beta v B^*}{H + P^*}\right)$.	$K > \dfrac{\mu}{\varepsilon a}$ and $\dfrac{\gamma v P^*}{H + P^*} > d$	Very complicated, but can be stable or unstable, depending on parameters.

3 Equilibrium Calculations

3.1 Set the Derivatives Equal to Zero and Factor

For the plant equation, factor out P^* and then collect the terms containing $1 - P^*/K$:

$$\begin{aligned} 0 &= rP^* \left(1 - \frac{P^*}{K}\right) - aP^*C^* + \beta \frac{vP^*B^*}{H + P^*} \left(1 - \frac{P^*}{K}\right) \\ &= P^* \left[r \left(1 - \frac{P^*}{K}\right) - aC^* + \frac{\beta v B^*}{H + P^*} \left(1 - \frac{P^*}{K}\right) \right] \\ &= P^* \left[\left(1 - \frac{P^*}{K}\right) \left(r + \frac{\beta v B^*}{H + P^*} \right) - aC^* \right]. \end{aligned}$$

For the caterpillar and bee equations, factor out C^* and B^*:

$$\begin{aligned} 0 &= \varepsilon a P^* C^* - \mu C^* = C^*(\varepsilon a P^* - \mu), \\ 0 &= \gamma \frac{vP^*B^*}{H + P^*} - dB^* - q(B^*)^2 \\ &= B^* \left(\frac{\gamma v P^*}{H + P^*} - d - qB^* \right). \end{aligned}$$

Thus, the equilibrium conditions are

$$0 = P^* \left[\left(1 - \frac{P^*}{K}\right) \left(r + \frac{\beta v B^*}{H + P^*} \right) - aC^* \right], \tag{2}$$

$$0 = C^*(\varepsilon a P^* - \mu), \tag{3}$$

$$0 = B^* \left(\frac{\gamma v P^*}{H + P^*} - d - qB^* \right). \tag{4}$$

Each product can be zero because its population factor is zero or because the remaining factor is zero. We examine these cases separately before dividing by any population.

3.2 Case 1: No Plants ($P^* = 0$)

Substitute $P^* = 0$ into Equation (3):

$$\begin{aligned} 0 &= C^*(\varepsilon a \cdot 0 - \mu) \\ &= -\mu C^*. \end{aligned}$$

Since $\mu > 0$, we obtain $C^* = 0$. Substitute $P^* = 0$ into Equation (4):

$$\begin{aligned} 0 &= B^* \left(\frac{\gamma v \cdot 0}{H + 0} - d - qB^* \right) \\ &= -B^*(d + qB^*). \end{aligned}$$

Therefore,

$$B^* = 0 \qquad \text{or} \qquad d + qB^* = 0 \quad \Longrightarrow \quad B^* = -\frac{d}{q}.$$

The second solution is negative, so it is not biologically meaningful. The plant equation is already zero because $P^* = 0$. Thus, the only nonnegative equilibrium without plants is

$$E_0 = (0, 0, 0). \tag{5}$$

This extinction equilibrium exists for all positive parameter values.

3.3 Case 2: Plants Present, No Caterpillars ($P^* > 0$, $C^* = 0$)

With $C^* = 0$, the caterpillar equation is automatically satisfied. Substitute $C^* = 0$ into Equation (2) and divide by $P^* > 0$:

$$0 = P^* \left(1 - \frac{P^*}{K}\right) \left(r + \frac{\beta v B^*}{H + P^*}\right),$$
$$0 = \left(1 - \frac{P^*}{K}\right) \left(r + \frac{\beta v B^*}{H + P^*}\right).$$

Because $r > 0$, $H + P^* > 0$, and $B^* \geq 0$,

$$r + \frac{\beta v B^*}{H + P^*} > 0.$$

Consequently, the other factor must be zero:

$$1 - \frac{P^*}{K} = 0,$$
$$\frac{P^*}{K} = 1,$$
$$P^* = K.$$

Substitute $P^* = K$ into Equation (4):

$$0 = B^* \left(\frac{\gamma v K}{H + K} - d - q B^*\right).$$

There are two possibilities.

3.3.1 No Bees: Plant-Only Equilibrium

Choose the first factor, $B^* = 0$. Together with $P^* = K$ and $C^* = 0$, this gives

$$E_P = (K, 0, 0). \tag{6}$$

This equilibrium exists for all positive parameter values.

3.3.2 Bees Present: Plant–Bee Equilibrium

If $B^* > 0$, divide the bee equation by B^* and solve:

$$\frac{\gamma v K}{H + K} - d - q B^* = 0,$$
$$q B^* = \frac{\gamma v K}{H + K} - d,$$
$$B^* = \frac{1}{q} \left(\frac{\gamma v K}{H + K} - d\right).$$

The plant–bee equilibrium is therefore

$$E_{PB} = \left(K,\, 0,\, \frac{1}{q} \left(\frac{\gamma v K}{H + K} - d\right)\right). \tag{7}$$

The condition for bees to be present is $\gamma v K/(H + K) > d$. At equality, $B^* = 0$ and this equilibrium coincides with E_P.

3.4 Case 3: Plants and Caterpillars Present ($P^* > 0$, $C^* > 0$)

Because $C^* > 0$, divide Equation (3) by C^*:

$$\begin{aligned} \varepsilon a P^* - \mu &= 0, \\ \varepsilon a P^* &= \mu, \\ P^* &= \frac{\mu}{\varepsilon a}. \end{aligned}$$

This fixes P^* whenever caterpillars are present.

Next, divide Equation (2) by $P^* > 0$ and solve for C^*:

$$\begin{aligned} \left(1 - \frac{P^*}{K}\right)\left(r + \frac{\beta v B^*}{H + P^*}\right) - aC^* &= 0, \\ aC^* &= \left(1 - \frac{P^*}{K}\right)\left(r + \frac{\beta v B^*}{H + P^*}\right), \\ C^* &= \frac{1}{a}\left(1 - \frac{P^*}{K}\right)\left(r + \frac{\beta v B^*}{H + P^*}\right). \end{aligned}$$

The bee equation becomes

$$0 = B^*\left(\frac{\gamma v P^*}{H + P^*} - d - qB^*\right),$$

which again gives two possibilities.

3.4.1 No Bees: Plant–Caterpillar Equilibrium

Set $B^* = 0$ in the expression for C^*:

$$\begin{aligned} C^* &= \frac{1}{a}\left(1 - \frac{P^*}{K}\right)\left(r + \frac{\beta v \cdot 0}{H + P^*}\right) \\ &= \frac{r}{a}\left(1 - \frac{P^*}{K}\right) \\ &= \frac{r}{a}\left(1 - \frac{\mu}{\varepsilon a K}\right). \end{aligned}$$

Thus, the plant–caterpillar equilibrium is

$$E_{PC} = \left(\frac{\mu}{\varepsilon a}, \frac{r}{a}\left(1 - \frac{\mu}{\varepsilon a K}\right), 0\right). \tag{8}$$

Because $r/a > 0$, caterpillars are strictly positive exactly when

$$\begin{aligned} 1 - \frac{\mu}{\varepsilon a K} &> 0, \\ \frac{\mu}{\varepsilon a K} &< 1, \\ \mu &< \varepsilon a K, \\ K &> \frac{\mu}{\varepsilon a} = P^*. \end{aligned}$$

At equality, $C^* = 0$ and E_{PC} coincides with E_P.

3.4.2 Bees Present: Three-Population Coexistence Equilibrium

If $B^* > 0$, divide the bee equation by B^* and solve:

$$\begin{aligned}
\frac{\gamma v P^*}{H + P^*} - d - qB^* &= 0, \\
qB^* &= \frac{\gamma v P^*}{H + P^*} - d, \\
B^* &= \frac{1}{q}\left(\frac{\gamma v P^*}{H + P^*} - d\right).
\end{aligned}$$

Substitute this expression into the formula for C^*:

$$\begin{aligned}
C^* &= \frac{1}{a}\left(1 - \frac{P^*}{K}\right)\left(r + \frac{\beta v}{H + P^*} B^*\right) \\
&= \frac{1}{a}\left(1 - \frac{P^*}{K}\right)\left[r + \frac{\beta v}{q(H + P^*)}\left(\frac{\gamma v P^*}{H + P^*} - d\right)\right].
\end{aligned}$$

Therefore, the three-population equilibrium is $E_{PCB} = (P^*, C^*, B^*)$, where

$$\begin{aligned}
C^* &= \frac{1}{a}\left(1 - \frac{P^*}{K}\right)\left[r + \frac{\beta v}{q(H + P^*)}\left(\frac{\gamma v P^*}{H + P^*} - d\right)\right], \\
B^* &= \frac{1}{q}\left(\frac{\gamma v P^*}{H + P^*} - d\right).
\end{aligned} \tag{9}$$

To determine when all three populations are positive, first require $B^* > 0$. Multiplying by the positive quantities q and $H + P^*$ gives

$$\begin{aligned}
\frac{\gamma v P^*}{H + P^*} - d &> 0, \\
\gamma v P^* - d(H + P^*) &> 0, \\
(\gamma v - d)P^* &> dH.
\end{aligned}$$

As in the plant–bee case, this requires $\gamma v > d$. Then

$$P^* > \frac{dH}{\gamma v - d}.$$

Once $B^* > 0$, the factor $r + \beta v B^*/(H + P^*)$ in C^* is positive. Since $a > 0$, we therefore have

$$\begin{aligned}
C^* > 0 \quad &\Longleftrightarrow \quad 1 - \frac{P^*}{K} > 0 \\
&\Longleftrightarrow \quad K > P^*.
\end{aligned}$$

Thus, all three populations are strictly positive exactly when

$$\gamma v > d, \qquad \frac{dH}{\gamma v - d} < \frac{\mu}{\varepsilon a} < K. \tag{10}$$

Equivalently, these conditions are

$$K > \frac{\mu}{\varepsilon a} \qquad \text{and} \qquad (\gamma v - d)\mu > dH\varepsilon a.$$

If $B^* = 0$, the coexistence formula reduces to E_{PC}. If $C^* = 0$ while $B^* > 0$, then $K = P^*$ and it reduces to E_{PB}. These boundary coincidences do not give additional distinct equilibria.

The cases above cover all possibilities. There are eight combinations of P^*, C^*, and B^*, where each population can be positive or equal to zero. The equilibrium cases we found cover five of these combinations. The remaining three have $P^* = 0$ and either $C^* > 0$, $B^* > 0$, or both. However, we showed that these cannot be equilibria: if $P^* = 0$, the equilibrium equations force $C^* = 0$ and $B^* = 0$. Thus, we have considered all possible nonnegative equilibria.

4 Stability

The Jacobian is

$$J(P,C,B) = \begin{pmatrix} J_{11}(P,C,B) & -aP & \dfrac{\beta v P}{H+P}\left(1-\dfrac{P}{K}\right) \\ \varepsilon a C & \varepsilon a P - \mu & 0 \\ \dfrac{\gamma v B H}{(H+P)^2} & 0 & \dfrac{\gamma v P}{H+P} - d - 2qB \end{pmatrix}. \tag{11}$$

The upper-left entry is

$$J_{11}(P,C,B) = r\left(1-\frac{2P}{K}\right) - aC + \frac{\beta v B\left[H(K-2P) - P^2\right]}{K(H+P)^2}.$$

4.1 Equilibrium $E_0 = (0,0,0)$

$$\begin{aligned}
&J_{11}(0,0,0) = r\left(1-\frac{2\cdot 0}{K}\right) - a\cdot 0 + 0 = r,\\
&J_{12}(0,0,0) = -a\cdot 0 = 0, \qquad J_{21}(0,0,0) = \varepsilon a\cdot 0 = 0,\\
&J_{13}(0,0,0) = \frac{\beta v\cdot 0}{H+0}\left(1-\frac{0}{K}\right) = 0,\\
&J_{22}(0,0,0) = \varepsilon a\cdot 0 - \mu = -\mu,\\
&J_{23}(0,0,0) = 0, \qquad J_{32}(0,0,0) = 0,\\
&J_{31}(0,0,0) = \frac{\gamma v\cdot 0\cdot H}{(H+0)^2} = 0,\\
&J_{33}(0,0,0) = \frac{\gamma v\cdot 0}{H+0} - d - 2q\cdot 0 = -d.
\end{aligned}$$

Therefore,

$$J(E_0) = J(0,0,0) = \begin{pmatrix} r & 0 & 0 \\ 0 & -\mu & 0 \\ 0 & 0 & -d \end{pmatrix}.$$

Thus,

$$\lambda_1 = r, \qquad \lambda_2 = -\mu, \qquad \lambda_3 = -d.$$

Since all parameters are positive, $\lambda_1 > 0$ while $\lambda_2, \lambda_3 < 0$. Thus, E_0 is an unstable saddle for all positive parameter values.

4.2 Equilibrium $E_P = (K, 0, 0)$

Substitute $P = K$, $C = 0$, and $B = 0$. Because $B = 0$, the bee-dependent term in J_{11} is zero:

$$\begin{aligned}
J_{11}(K,0,0) &= r\left(1 - \frac{2K}{K}\right) - a \cdot 0 + 0 \\
&= r(1-2) = -r, \\
J_{12}(K,0,0) &= -aK, \\
J_{13}(K,0,0) &= \frac{\beta v K}{H+K}\left(1 - \frac{K}{K}\right) = \frac{\beta v K}{H+K} \cdot 0 = 0, \\
J_{21}(K,0,0) &= \varepsilon a \cdot 0 = 0, \\
J_{22}(K,0,0) &= \varepsilon a K - \mu, \\
J_{23}(K,0,0) &= 0, \qquad J_{32}(K,0,0) = 0, \\
J_{31}(K,0,0) &= \frac{\gamma v \cdot 0 \cdot H}{(H+K)^2} = 0, \\
J_{33}(K,0,0) &= \frac{\gamma v K}{H+K} - d - 2q \cdot 0 = \frac{\gamma v K}{H+K} - d.
\end{aligned}$$

Therefore,

$$J(E_P) = J(K,0,0) = \begin{pmatrix} -r & -aK & 0 \\ 0 & \varepsilon a K - \mu & 0 \\ 0 & 0 & \dfrac{\gamma v K}{H+K} - d \end{pmatrix}.$$

The characteristic polynomial is:

$$\begin{aligned}
0 &= \det(\lambda I - J(E_P)) \\
&= (\lambda + r)(\lambda + \mu - \varepsilon a K)\left(\lambda + d - \frac{\gamma v K}{H+K}\right).
\end{aligned}$$

Setting each factor equal to zero gives

$$\lambda_1 = -r, \qquad \lambda_2 = \varepsilon a K - \mu, \qquad \lambda_3 = \frac{\gamma v K}{H+K} - d.$$

$\lambda_1 = -r < 0$ automatically. For the second eigenvalue,

$$\begin{aligned}
\lambda_2 < 0 \quad &\iff \quad \varepsilon a K - \mu < 0 \\
&\iff \quad \varepsilon a K < \mu \\
&\iff \quad K < \frac{\mu}{\varepsilon a}.
\end{aligned}$$

For the third eigenvalue,

$$\begin{aligned}
\lambda_3 < 0 \quad &\iff \quad \frac{\gamma v K}{H+K} - d < 0 \\
&\iff \quad \frac{\gamma v K}{H+K} < d.
\end{aligned}$$

Thus, E_P is locally asymptotically stable when both

$$K < \frac{\mu}{\varepsilon a} \qquad \text{and} \qquad \frac{\gamma v K}{H+K} < d.$$

4.3 Equilibrium E_{PB}

$$\begin{aligned} J_{11}(K,0,B^*) &= r\left(1-\frac{2K}{K}\right) - a\cdot 0 + \frac{\beta v B^*\left[H(K-2K)-K^2\right]}{K(H+K)^2} \\ &= -r + \frac{\beta v B^*(-HK-K^2)}{K(H+K)^2} \\ &= -r - \frac{\beta v B^* K(H+K)}{K(H+K)^2} \\ &= -r - \frac{\beta v B^*}{H+K}. \end{aligned}$$

$$\begin{aligned} J_{12}(K,0,B^*) &= -aK, \\ J_{13}(K,0,B^*) &= \frac{\beta v K}{H+K}\left(1-\frac{K}{K}\right) = 0, \\ J_{21}(K,0,B^*) &= \varepsilon a\cdot 0 = 0, \\ J_{22}(K,0,B^*) &= \varepsilon a K - \mu, \\ J_{23}(K,0,B^*) &= 0, \qquad J_{32}(K,0,B^*) = 0, \\ J_{31}(K,0,B^*) &= \frac{\gamma v B^* H}{(H+K)^2}, \\ J_{33}(K,0,B^*) &= \frac{\gamma v K}{H+K} - d - 2qB^* \\ &= qB^* - 2qB^* = -qB^*. \end{aligned}$$

Thus,

$$J(E_{PB}) = \begin{pmatrix} -r - \dfrac{\beta v B^*}{H+K} & -aK & 0 \\ 0 & \varepsilon a K - \mu & 0 \\ \dfrac{\gamma v B^* H}{(H+K)^2} & 0 & -qB^* \end{pmatrix}.$$

Expanding the determinant along the second row gives

$$\begin{aligned} 0 &= \det(\lambda I - J(E_{PB})) \\ &= (\lambda + \mu - \varepsilon a K)\begin{vmatrix} \lambda + r + \dfrac{\beta v B^*}{H+K} & 0 \\ -\dfrac{\gamma v B^* H}{(H+K)^2} & \lambda + qB^* \end{vmatrix} \\ &= (\lambda + \mu - \varepsilon a K)\left(\lambda + r + \frac{\beta v B^*}{H+K}\right)(\lambda + qB^*). \end{aligned}$$

Setting each factor equal to zero gives

$$\begin{aligned} \lambda_1 &= -r - \frac{\beta v B^*}{H+K}, \\ \lambda_2 &= \varepsilon a K - \mu, \\ \lambda_3 &= -qB^*. \end{aligned}$$

Since the parameters and B^* are positive, $\lambda_1 < 0$ and $\lambda_3 < 0$. For the remaining eigenvalue,

$$\begin{aligned} \lambda_2 < 0 \quad &\iff \quad \varepsilon a K - \mu < 0 \\ &\iff \quad \varepsilon a K < \mu \\ &\iff \quad K < \frac{\mu}{\varepsilon a}. \end{aligned}$$

Therefore, the plant–bee equilibrium with $B^* > 0$ is locally asymptotically stable when

$$\frac{\gamma v K}{H + K} > d \qquad \text{and} \qquad K < \frac{\mu}{\varepsilon a}.$$